

MUHAMMAD HAIDER ZAIDI

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🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python (Fluent), C++, SQL, Javascript

Deep Learning: PyTorch (ML Training Loops), TensorFlow, Computer Vision, Neural Networks, Generative AI

MLOps & AI Workflows: Docker, Kubernetes, AWS SageMaker, FastAPI, CI/CD, GPU Optimization, ML Infrastructure, AI Tooling

Data Engineering: PySpark, Airflow, ETL Pipelines, Data Analysis Pipelines, Data Processing

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated deep learning models, improving model performance and accelerating ML training efficiency for metrological applications.
- Explored novel architectures and physics-informed hybrid loss functions to enhance model accuracy and stability in research workflows.
- Developed data analysis pipelines and interactive dashboards for robust model performance monitoring and debugging.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Improved vitiligo segmentation accuracy by 94% and accelerated inference speed by 5x using AI algorithms, deployed with Docker and FastAPI.
- Developed a real-time computer vision system leveraging TensorRT for GPU optimization, reducing compute cost and time by 6x.
- Managed robust MLOps pipelines on AWS SageMaker, addressed data drift, and maintained Kubernetes infrastructure for scalable ML deployments.
- Designed and developed AI-driven Proof-of-Concepts (POCs) utilizing YOLO, few-shot learning, and CNNs for client demonstrations.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Developed full-stack web applications providing AI-based services, integrating machine learning models into user-facing interfaces.
- Engineered an AI chatbot application leveraging the OpenAI API and advanced LLM tools/frameworks, contributing to AI-driven workflows.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Developed ETL and data analysis pipelines for structured and unstructured data processing, supporting client analytics.
- Implemented automated data ingestion and cleaning workflows, enhancing data quality and preparing data for AI applications.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Architected a weighted-probability ensemble integrating DPFunc and DeepGOPlus, leveraging AlphaFold for feature extraction and taxonomic bias mitigation.
- Increased model accuracy by 4% through custom feature engineering across MF, CC, and BP namespaces, demonstrating strong ML fundamentals.

Razanalytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Developed a real-time Bitcoin price forecasting system, employing advanced machine learning models for predictive analytics.
- Implemented data analysis pipelines for continuous model evaluation and performance monitoring in a dynamic environment.

Saaz: AI Self-Playing Chitralli Sitar

2023

Signal Processing, Music Generation

- Co-developed the world's first AI-driven self-playing Chitralli Sitar, integrating real-time signal processing with music generation LLMs.
- Designed a real-time signal processing system integrating temporal dependencies for music generation, leveraging advanced deep learning techniques.
- Implemented transformer-based models for sequence prediction in audio generation tasks, demonstrating expertise in generative AI.

Hugging Face

Open-Source

Contributor

- Co-authored the Zero-shot Image Classification technical documentation and implementation page on the Hugging Face Hub.
- Contributed to open-source initiatives by documenting and implementing advanced computer vision techniques.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Specialized in deep learning for computer vision and spatio-temporal modeling
- Relevant coursework: Advanced Neural Networks, Computer Vision, Autonomous Systems

National University of Computer and Emerging Sciences Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Dean's List of Honor
- Specialized in Machine Learning and Computer Vision with focus on time-series analysis

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Datacamp:** Computer Vision in Python